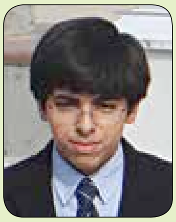
COMMUNITY CREATIONS



About the creator

Anubhav Wadhwa (aged 16) is a technology entrepreneur. He founded TechAPTO, trends on internet and Tyrelessly. He believes in contributing towards the fulfilment of sustainable development goals through corroborative participation.

About his creation

At the age of 13, Anubhav established TechAPTO and in the year 2013, the venture was selected from 600 startups and made it to the top 40 across India, which translated to top 10 in Delhi and NCR. In 2015, Anubhav founded www.tyrelessly.com from his internal accruals originating from his 3 year old venture, TechAPTO.

PIR Motion Sensor

Talk about detecting motion and here we have our ₹100 sensor that will detect motion for you. As the name suggests, PIR stands for Passive Infrared and it works by detecting differences in Infrared light emitted by objects. As soon as it notices any differences in infrared levels, it registers it as motion. It comes with three pins VCC (5 volts), GND and OUT.

The connection is again pretty simple. Connect the VCC pin to the '5 Volt' port on the Arduino and the 'GND' pin goes into the GND port. In fact, these are some conventions that you must get used to by now because you'll encounter these two pins in almost every sensor. And the other pins are generally connected to the Digital pins and in the PIR sensor, connect the 'OUT' pin to the Digital Pin no. 8 (you can choose other ports too, it's totally your choice, just make sure to make relevant changes in the code).

Open up the Arduino IDE and enter the code as given on http://pastebin.com/AE35ac9N. This code will turn ON or OFF the onboard LED light if any motion is detected. So, edit that as per your program's requirement and you'll start using the sensor.

Temperature and Humidity Sensor

Almost every variant must have at least three pins, each for 5 volts, ground and signal respectively.

They work by calculating the electrical resistance between the two electrodes that is present beneath it but you need not worry about that, rather, let's get familiar with its connections. Chances are that instead of 'VCC' or 'GND', you may notice that it has plus and minus signs marked respectively. 'SIGNAL' pin would be marked with an 'S'. This time, try to connect them with Arduino on your own (after all, you won't blow it up). For now, we'll connect the SIGNAL pin to the DIGITAL pin no. 7.

Fortunately, we've got a library function for this sensor which you can download from http://bit.ly/1PPAiZZ and extract it into the Documents → Arduino → libraries folder. Get back to Arduino IDE and get the sample code from http://pastebin.com/tRaThA9h. Hit the upload button and open up the serial monitor and you'll see the temperature and the humidity coming up.

Relay

This exactly isn't a sensor but is useful as it acts as an electric switch and you would be using it often. It generally comes in multiple channels which translate into "how many individual switches you want?" Buying a '5 Volt 4 Channel relay' would be ideal for most of you. It will have 4 pins, 1 for each switch and 2 more pins for GND and 5 volts. On the other side, the relay's switch will contain 3 ports. C stands for Common, NO stands for Normally Open and NC means Normally closed. One end of the wire will go in the 'Common' port and the other in either of the remaining two ports, NC or NO, depending on your usage.

Library file for this one would make no sense since all it needs to do is to accept either HIGH (1) or LOW (0) commands from Arduino. I've made a small program that will demonstrate the usage of relay using the temperature sensor. You can copy the code from http://pastebin.com/cW15YBU5 and as you can see, I've used the functions from Temperature sensor, thus, if the temperature is above 20 it will turn ON the appliance (if it doesn't then you'll need to put the 'second' wire into NO port).

And guess what, you just used two different sensors to build a project that is intuitive enough to take surrounding's temperature and 'react' accordingly. Likewise, there are other tons of available sensors which you can combine together and create something more awesome. Happy DIY!